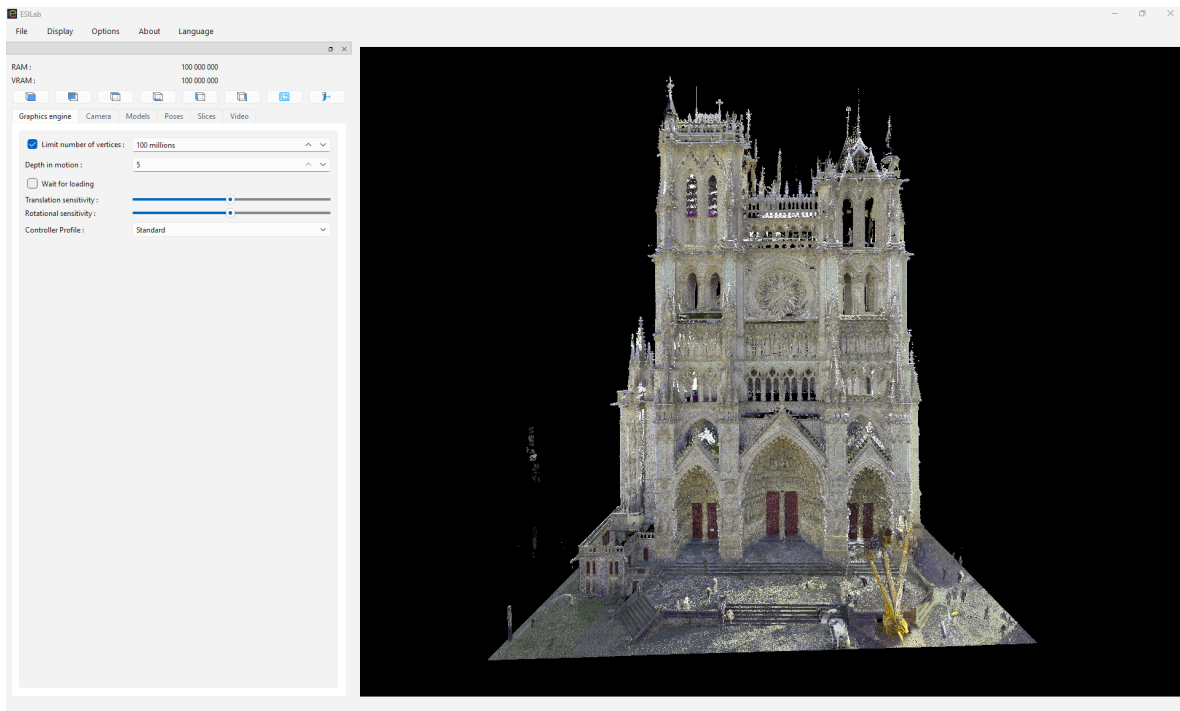


ESILab: Enhanced Scan Immersion Lab.

Version 2.8.1

June 15, 2026



Contents

1	Introduction	2
2	Installation Windows	3
3	Usage	3
4	Features	3
4.1	Menu bar	4
4.2	Visualization Window	6
4.3	Parameters Window - Header	6
4.4	Parameters Window - Graphics Engine tab	7
4.5	Parameters Window - Camera tab	7
4.6	Parameters Window - Models tab	10
4.7	Parameters Window - Poses tab	12
4.8	Parameters Window - Slices tab	12
4.9	Parameters Window - Video tab	13

1 Introduction

This article presents ESILab, a free software application primarily dedicated to the visualization of very large point clouds.

ESILab was developed in response to the internal needs of our research team and has been used to process point clouds of major large-scale monuments in France and abroad, including the cathedrals of Amiens, Beauvais, and Paris. Around these 3D models, we have collaborated on numerous projects aimed at studying, understanding, diagnosing, and producing deliverables that can be effectively used by archaeologists, art historians, and architects.

The software was designed by a team of researchers in computer vision and robotics to address the many requirements of archaeologists, architects, and architectural historians.

ESILab is capable of loading 3D models containing several billion points within a few seconds. It also allows multiple point clouds to be loaded simultaneously and accurately overlaid. These performances are made possible through the structuring of point cloud data and the extensive configuration options available for the rendering engine.

Direct access to the transformation matrices of each point cloud, either through text files or via the graphical user interface, enables users to specify the pose of individual datasets. This feature makes the alignment, superimposition, and comparison of multiple models both straightforward and precise.

ESILab supports perspective, orthographic, and equirectangular rendering modes.

These camera models can be configured either through the user interface or via text-based configuration files.

Users can access and modify horizontal and vertical focal lengths, image dimensions, and their equivalent intrinsic camera parameters commonly used in the computer vision and robotics communities.

Main features of ESILab:

- Smooth navigation through extremely large point clouds.
- A comprehensive set of tools for data exploitation, analysis, and interpretation, including model comparison, overlay, cross-section extraction, video generation, and more.
- Extensive camera configuration options covering standard requirements (field of view, focal length, image size, etc.) as well as advanced computer vision parameters (camera intrinsic parameters).
- In addition to conventional perspective and orthographic cameras, ESILab provides an equirectangular mode, enabling renderings equivalent to those produced by spherical cameras commonly used for virtual tours.
- Numerous rendering engine settings with per-model customization capabilities.
- Direct access to the positions and orientations of 3D models.
- Axis inversion capabilities, useful for applying symmetry transformations to models.

2 Installation Windows

- Download the Windows installer [Esilab.Setup.exe](#) or visit the official [GitHub page](#).
- Run the installer from your Downloads folder. If a Windows security warning appears, click **More info**, then **Run anyway**.
- Follow the installation wizard.

3 Usage

ESILab allows you to visualize and navigate through point clouds based on point clouds in **OCTI** format. This format is generated from **PTS** files using the [3DModelConverter](#) software, which must be used in order to properly use ESILab.

- Click **File/Open Model** and select an **OCTI** file. Make sure that the file **name_LO.txt** is in the same folder as the **name.octi** file and that both share the same base name **name**.
- Use ESILab features to navigate through the point cloud, take screenshots, create cross-sections, generate videos, etc... See the section below for more details.

4 Features

After starting ESILab, the start page appears (see Fig.1).

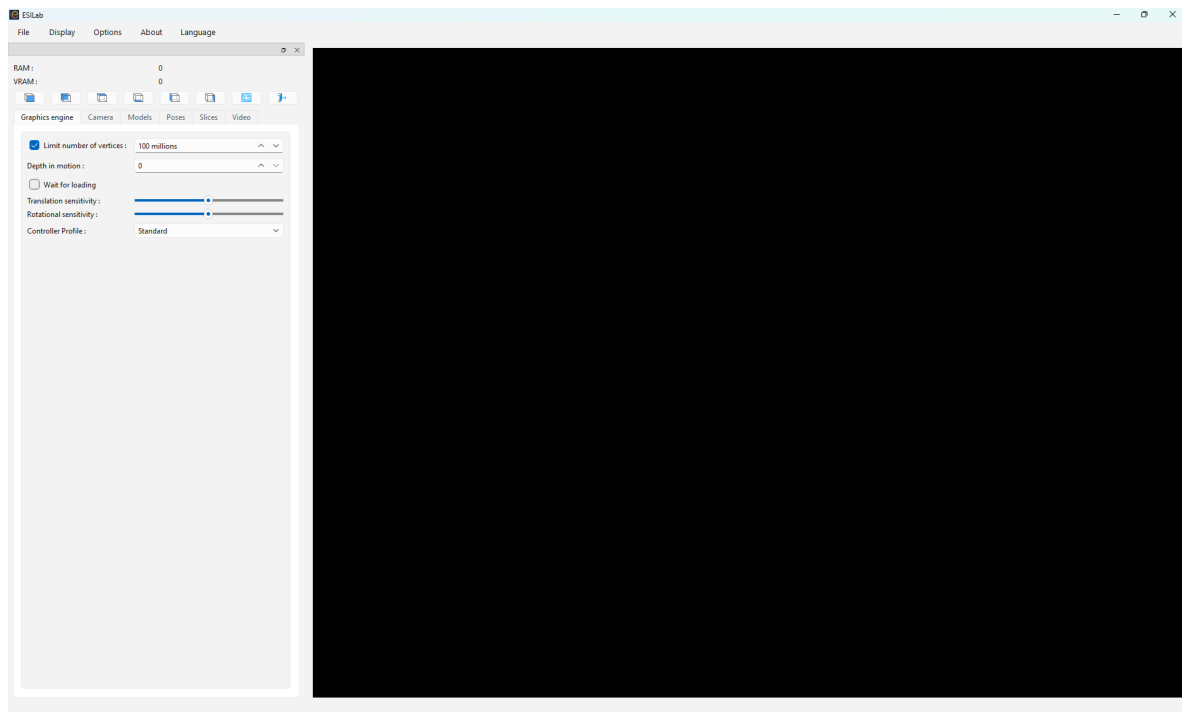


Figure 1: ESILab start page

The ESILab interface is mainly composed of (see Fig.2):

- a menu bar (highlighted in purple),
- a visualization window (highlighted in yellow),
- a parameters window (highlighted in green),

- the header of the parameters window (highlighted in blue),
- the tabs of the parameters window (highlighted in pink).

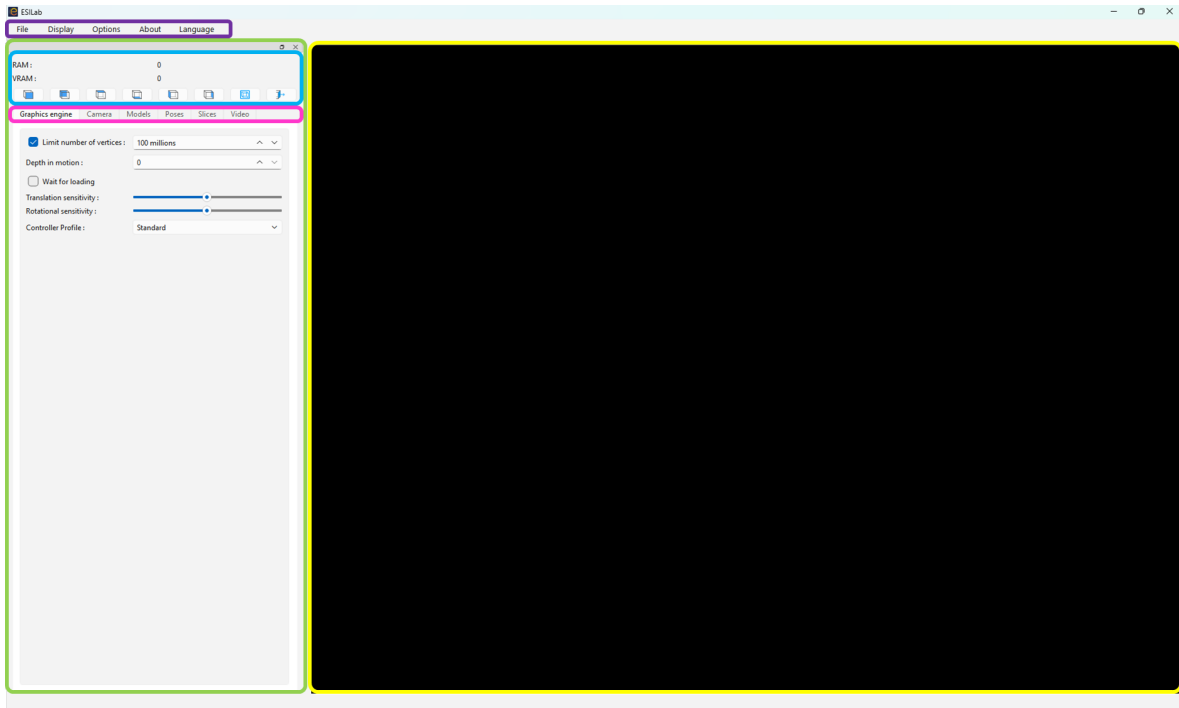


Figure 2: ESILab start page with highlighted menus

The functionalities accessible through these different parts of the interface are described in the sections below.

4.1 Menu bar

The **File** menu allows you to:

- Load a point cloud in .OCTI format using **Open Model**. Make sure that the file **name_LO.txt** is in the same folder as the **name.octi** file and that both share the same base name **name**.
- Load camera settings from a .INI file using **Open Camera**.
- Save the current camera settings to a .INI file using **Record Camera**.

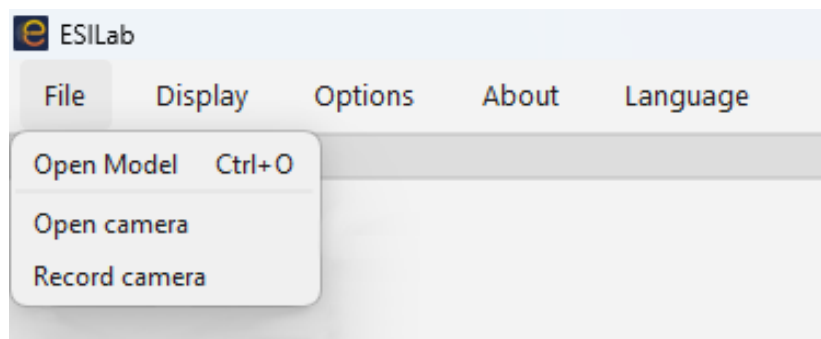


Figure 3: File menu

The **Display** menu allows you to:

- Show or hide the parameters window using **Parameters**.
- Show or hide the visualization window in full screen using **Full Screen**.

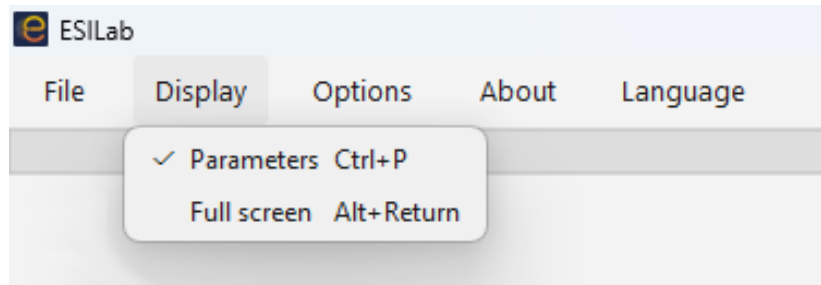


Figure 4: Display menu

The **Options** menu allows you to:

- Save a local image of the current render of the visualization window using **Capture**.
- Change the background color of the visualization window using **Background Color**.
- Save the current camera pose in the scene using **Save Start Pose**.

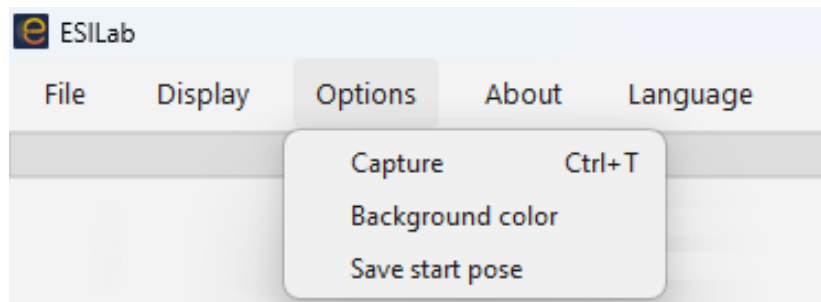


Figure 5: Options menu

The **About** menu displays the application version information and additional details.

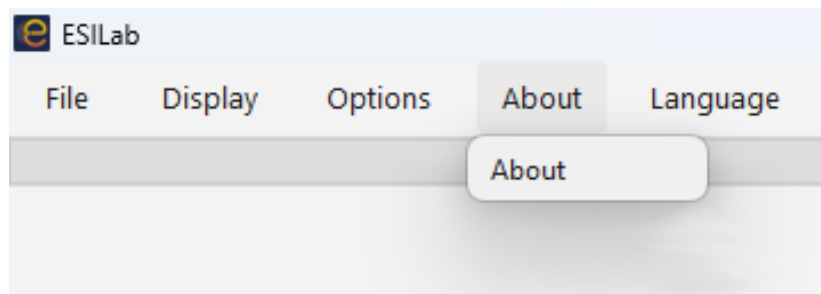


Figure 6: About menu

In the menu bar, the **Language** menu allows you to switch the software language between English and French.

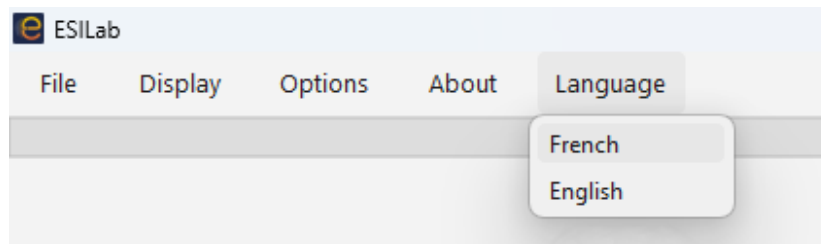


Figure 7: Language menu

4.2 Visualization Window

The visualization window allows you to display one or more point clouds and navigate within the scene. To navigate, use the keyboard and mouse:

- left click mouse - camera rotation on x, y and z axis
- right click mouse - camera translation on x and y axis
- mouse wheel mouse - camera translation on z axis
- up/down arrows keys - camera translation on z axis
- left/right arrows keys - camera translation on x axis
- a/e keys - camera translation on y axis
- z/s keys - camera rotation on x axis
- q/d keys - camera rotation on y axis

It is also possible to move the virtual camera using a game controller connected to your PC.

4.3 Parameters Window - Header

The top of the parameters window header shows the number of vertex loaded into your machine's RAM and virtual RAM.

RAM :	1 000 000
VRAM :	1 000 000

Figure 8: Parameters Window - Header's top

The bottom of the parameters window header allows you to quickly move the camera within the scene relative to the loaded and selected point cloud model in the Model tab:

- front view,
- back view,
- top view,
- bottom view,
- left view,
- right view,
- directed toward the point cloud's center of gravity with a zoomed view,
- centered and facing a plane defined by selecting three points from the point cloud.



Figure 9: Parameters Window - Header's bottom

4.4 Parameters Window - Graphics Engine tab

The **Graphics Engine** tab allows you to:

- Set the maximum number of vertices displayed in the visualization window (in millions) using **Limit Number of Vertices**.
- Define the depth level of point cloud models displayed while the camera is moving through the scene using **Depth in Motion**.
- Choose whether to wait until all point cloud data has been loaded into RAM and virtual RAM before displaying the new render using **Wait for Loading**.
- Adjust translation sensitivity using **Translation Sensitivity**.
- Adjust rotation sensitivity using **Rotation Sensitivity**.
- Select the controller profile:
 - **Standard**
 - * Left joystick: translations along the X and Z axes.
 - * Right joystick: rotations around the X and Y axes.
 - * L1/R1: rotations around the Z axis.
 - * L2/R2: translations along the Y axis.
 - **Alternative**
 - * Left joystick: rotations around the X and Y axes.
 - * Right joystick: translations along the X and Z axes.
 - * Up/Down arrows: translations along the Y axis.

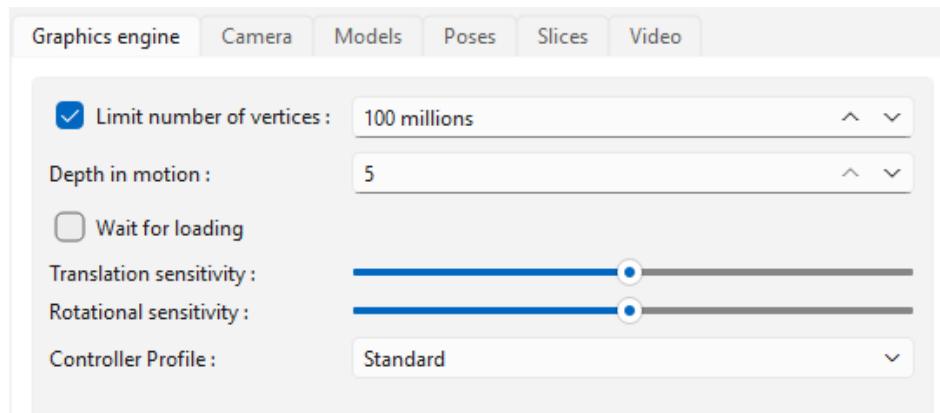


Figure 10: Parameters Window - Graphics Engine tab

4.5 Parameters Window - Camera tab

The **Camera** tab allows to configure the parameters of the virtual camera in the scene:

- The **wMc - Camera Position in World** section allows to:

- View / edit the camera position and rotation in **Position** and **Rotation** in the world coordinate system (reference frame of the virtual scene), along and around the X, Y, and Z axes.
- View / edit the camera rotations using **Yaw Pitch Roll**, corresponding to rotations around the X, Y, and Z axes.
- View the camera translation position along the X, Y, and Z axes in the coordinate system of the origin of the loaded and selected model in the Model tab with **Target**.
- The **Camera Settings** section allows you to:
 - Define whether the camera image plane **Length** and **Width** are set manually or automatically using the screen resolution via **Native Resolution**.
 - Modify the image plane **Length**, **Width**, field of view **FOV**, and horizontal field of view **HFOV** of the virtual camera.
 - Set the virtual camera depth range using **Near** and **Far**.
 - Define whether rotation around the camera optical axis is locked or not using **Vertical Axis**.
 - Define whether translation along the camera Y-axis is locked or not using **On the Ground**.
 - Set the antialiasing resolution between 0 and 32 using **Multi-sampling**.
- The **Projection** section allows you to choose the virtual camera rendering mode: **Perspective**, **Orthographic**, or **Equirectangular**.

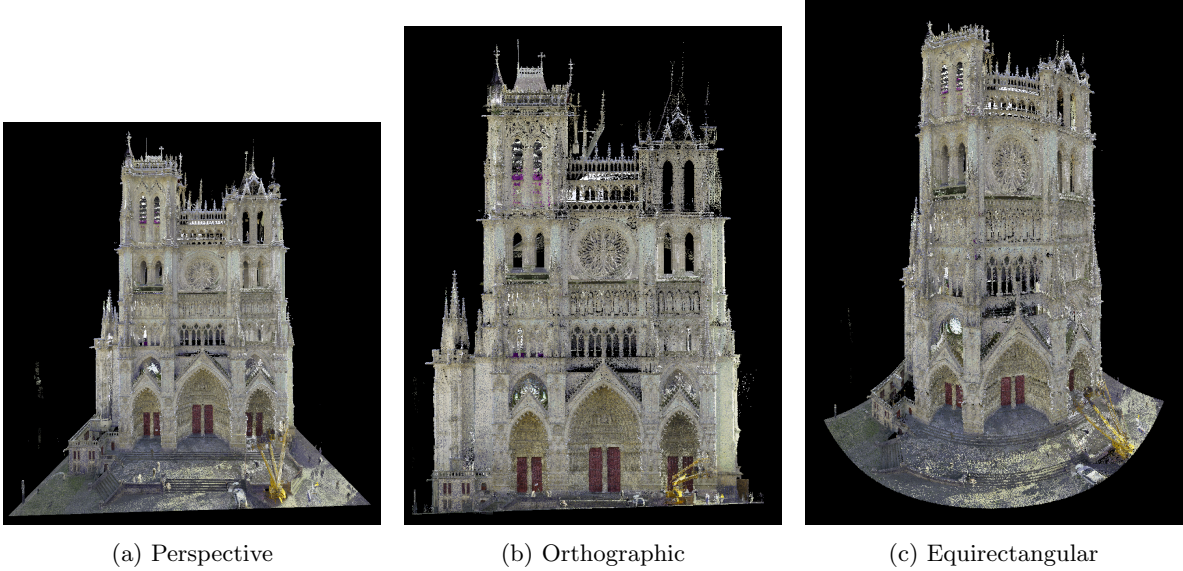


Figure 11: Renderings of the different projection modes

- The **Intrinsic Parameters** section allows you to view or edit the following parameters:

$$\alpha_u, \alpha_v, k_u, k_v, u_0, v_0$$

Graphics engine
Camera
Models
Poses
Slices
Video

wMc - Camera Position in World

Position :	-1,649500	-32,063000	56,649094
Rotation :	0,000000	0,000000	0,000000
Yaw Pitch Roll :	0,000000	0,000000	0,000000
Target :	-1,649500	-32,063000	55,649094

Camera Settings

☒ Native resolution

Length :	1920	^ v
Width :	1080	^ v
FOV :	90,00	^ v
HFOV :	103,26	^ v
Near :	0,10	^ v
Far :	1000,00	^ v

☒ Vertical Axis
☐ On the ground

Multi-sampling:	0	^ v
-----------------	---	-----

Projection

☒ Perspective

☐ Orthographic

☐ Equirectangular

Intrinsic parameters

α_u :	528,50	^ v
α_v :	528,50	^ v
k_u :	30,00	^ v
k_v :	30,00	^ v
u_0 :	667,50	^ v
v_0 :	528,50	^ v

Figure 12: Parameters Window - Camera tab

4.6 Parameters Window - Models tab

The **Models** tab allows you to:

- View and select point cloud models loaded into the software.
- For each selected model, you can:
 - View the number of **Vertex**.
 - Show or hide the point cloud in the visualization window using **Visible**.
 - Enable or disable and adjust point opacity using **Opacity**.
 - Select the color rendering mode using **Blend** when **Opacity** is enabled.
 - Enable or disable the display of point **Intensity**.
 - Adjust the distance scale between points using **Scale**.
 - Enable or disable and set the size of rendered points using **Dots Size**.
 - View / edit the model position in translation and rotation in the world coordinate system (reference frame of the virtual scene) along and around the X, Y, and Z axes using the **wMo** section.
 - Invert the model point positions along the **X**, **Y**, and **Z** axes.
 - Display AABB (Axis-Aligned Bounding Boxes) using **AABB**, and visualize:
 - * the point at minimum coordinates using **Min**,
 - * the point at maximum coordinates using **Max**,
 - * the center of gravity using **Gravity**,
 - View / edit the octree depth displayed in the visualization window using **Octrees Depth**.
 - Enable or disable and set a uniform color for all points in the point cloud using **Color**.
 - Close the selected model using **Close Model**.

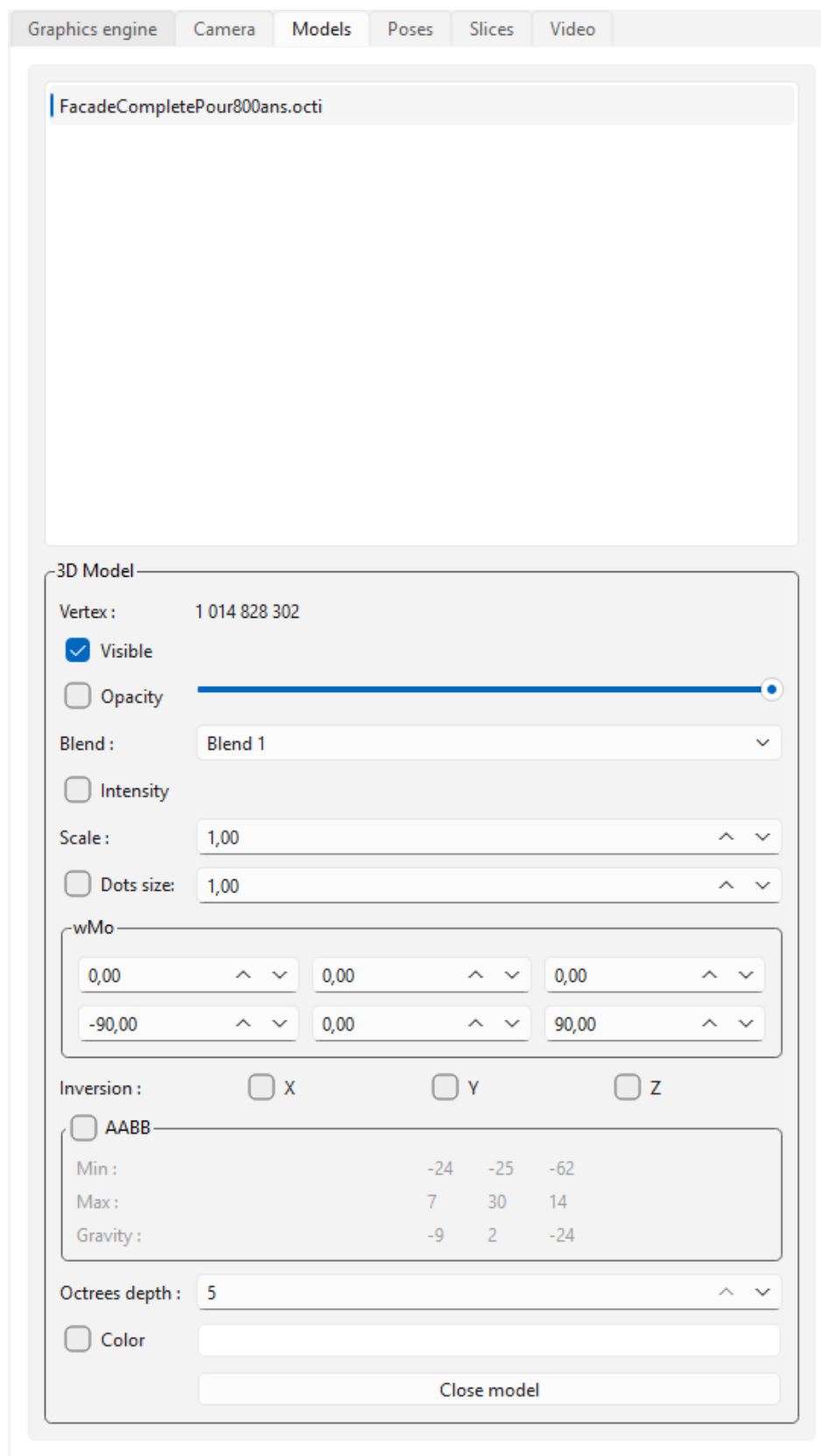


Figure 13: Parameters Window - Model tab

4.7 Parameters Window - Poses tab

The **Poses** tab allows you to:

- Save the current camera pose in the world coordinate system using **Add Current Pose**. This pose is stored in the **name.ini** file associated with the selected **name.octi** model in the Model tab. The **INI** file is automatically loaded when the corresponding **OCTI** model is opened in ESILab.
- Automatically move the camera to a saved pose by clicking on the corresponding entry in the pose list.
- Delete the selected pose using **Delete**.

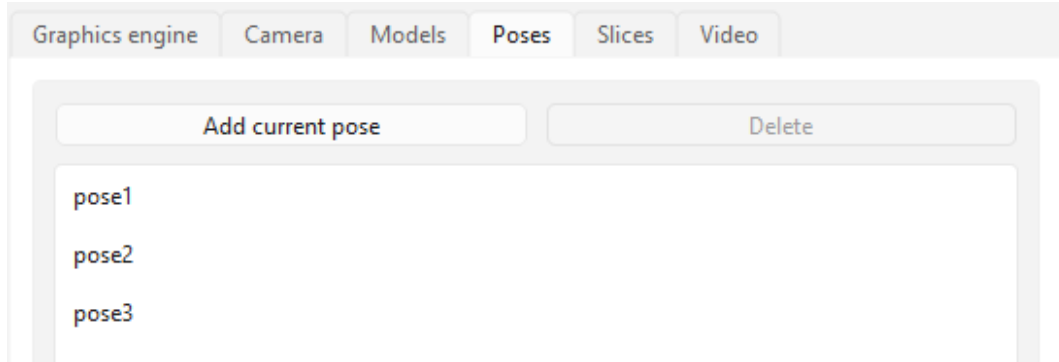


Figure 14: Parameters Window - Poses tab

4.8 Parameters Window - Slices tab

The **Slices** tab allows you to:

- View / edit the thickness of each slice using **Thickness**.
- View / edit the step between slices using **Step**.
- View / edit the distance between the first slice and the camera using **Start**. Click **Auto** to let ESILab automatically compute this distance based on the selected point cloud in the Model tab.
- View / edit the distance between the last slice and the camera using **Stop**. Click **Auto** to let ESILab automatically compute this distance based on the selected point cloud in the Model tab.
- Reverse the slicing order using **Reverse**.
- View / edit the display time of each slice during the test using **Test Times**.
- Run a slicing visualization test using **Test**.
- Save images of each slice using **Let's Start!**. Each slice is saved as a PNG file in a destination folder.
- View the distance between the current slice and the camera using **Slicing Position**.

For high-quality renders, it is recommended to follow the procedure below:

1. Select the **Orthographic** camera projection in the **Camera** tab.
2. Position the camera in the scene.
3. Click **Auto** for the **Start** position.
4. Click **Auto** for the **Stop** position.

5. Click **Test**.
6. Click **Let's Start!** if the test result is satisfactory.

Graphics engine Camera Models Poses Slices Video

Thickness : 1,000 ^ v

Step : 0,100 ^ v

Start : 0,100 ^ v Auto

Stop : 100,000 ^ v Auto

☐ Reverse

Test times : 10,00 ms ^ v

Test

Let's start!

Slicing position :

Figure 15: Parameters Window - Slices tab

4.9 Parameters Window - Video tab

The **Video** tab allows you to:

- Load all video generation parameters using **Load** from an **INI** file.
- Save all current video generation parameters using **Save** to an **INI** destination file.
- View / edit the number of frames per second for the final video rendering using **Frames per Second**.
- Save the current camera pose in the world coordinate system using **Add Pose**.
- Delete the selected pose using **Delete**.
- Change the order of poses using **Up** and **Down**.
- For each selected pose:
 - View / edit the camera transition time from the previous pose to the selected pose using **Time**.
 - View / edit the camera field of view using **FOV**.
 - View / edit the virtual camera depth range using **Near** and **Far**.
 - Define whether rotation around the camera optical axis is fixed using **Vertical Axis**.
 - Enable yaw-pitch-roll based rotation control using **Use Yaw Pitch Roll**.
 - Run a test preview of the final video using **Test**.
 - Run a test preview of the transition between the previous pose and the selected pose using **Test Transition**.
 - Export the final video in **MP4** format as well as all frame images in **PNG** format using **Capture**. A `namevideoCameraParameters.ini` file and a `namevideoEngineParameters.ini` file are also automatically generated. These files contain the settings from the **Graphics Engine** and **Camera** tabs used to generate the video.

Graphics engine
Camera
Models
Poses
Slices
Video

Video capture

Load

Save

Frames per second:
25
^
v

Add pose

Delete pose

Up

Down

(-1.6495 ; -32.063 ; 56.6491) (0 ; 0 ; 0)

(-1.6495 ; -32.063 ; 71.0491) (0 ; 0 ; 0)

(-0.0495001 ; -7.263 ; 96.2491) (0 ; 0 ; 0)

Pose

Time : 1000 ms ^ v

FOV : 90,00 ^ v

Near : 0,10 ^ v

Far : 1000,00 ^ v

Options

☐ Vertical Axis
☐ Use Yaw Pitch Roll

Test

Test transition

Capture

Figure 16: Parameters Window - Video tab